The Purification and Properties of Glutathione Peroxidase of Erythrocytes*

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Previous studies (1, 2) have shown the presence of a peroxidase in erythrocytes which catalyzes the breakdown of hydrogen peroxide with reduced glutathione serving as a hydrogen donor. The significance of this enzyme in the protection of hemoglobin from oxidative breakdown has been shown previously (2). The erythrocyte reduced glutathione peroxidase has now been purified by column chromatography on a modified cellulose anion exchange column. The peroxidase activity has been separated completely from the catalase activity, confirming the previous conclusion that the peroxidase and catalase activities of the red cell could be attributed to two distinct enzymes. Although the enzyme in impure preparations is relatively stable, purification results in a markedly increased lability.

In addition, some systematic studies have been carried out with this purified enzyme preparation to determine the enzyme specificity in regard to hydrogen donors. Plant peroxidases, in general, have a relatively low degree of specificity since a wide variety of compounds, including organic amines, aromatic diols, and sulfhydryl compounds, will serve as hydrogen donors (3–6). In contrast to this, certain animal peroxidases, such as tryptophan peroxidase, show considerable substrate specificity (4). Since reduced glutathione peroxidase differs from most other peroxidase enzymes in that cyanide and azide fail to inhibit its activity (1), it appears likely that there might be differences in specificity for hydrogen donors. In the present study, the enzyme specificity of reduced glutathione peroxidase has been studied from several different aspects. First, we have studied the nature of the hydrogen donor which is necessary for the reduction of hydrogen peroxide in the presence of peroxidase. In addition, other hydrogen donors have been utilized in place of ascorbic acid in the reaction with oxyhemoglobin to yield hydrogen peroxide, and the ability of the peroxidase-reduced glutathione system to protect hemoglobin from oxidation under these conditions has been determined.

EXPERIMENTAL

Materials—Crystalline rat oxyhemoglobin was prepared by the method of Heidelberger (7) after slight modification to adapt the procedure for use with rat blood. Solutions of cysteinylglycine (Schwarz Laboratories, Inc.), ergothioneine (California Foundation for Biochemical Research), thiomalate, cysteine, GSH, ascorbic acid, sodium azide, epinephrine, acetylphenylhydrazine and hydrogen peroxide were prepared fresh as needed for the various experimental procedures. The hydrogen peroxide concentration was determined by titration with a standardized permanganate solution. Horseradish peroxidase was obtained from Nutritional Biochemicals Corporation. Diethylaminoethyl cellulose[1] was prepared from Solka-Floc as described by Peterson and Sober (8). Titration curves of the lyophilized preparation showed the presence of 0.9 m.eq. of ionizing groups per gm. of dry material.

Methods—All spectrophotometric measurements were made with a model DU Beckman spectrophotometer. Optical density readings at 270 mμ were used as a routine measure of the protein concentration in eluent fractions. The optical density readings at 270 mμ of a solution of the purified enzyme preparation were correlated with the protein nitrogen content of this same sample by a nitrogen analysis carried out by the Kjeldahl procedure (9). Catalase activity was measured by the method of Feinstein (10).

Peroxidase Assay Procedure 1—In this procedure the ability of the peroxidase-GSH system to prevent the oxidative breakdown of hemoglobin is utilized as a measure of peroxidase activity. Each sample contained 2.4 ml. of hemoglobin (crystalline rat oxyhemoglobin, 0.125 gm. per 100 ml., in an isotonic Krebs-Ringer phosphate buffer, pH 7.3), 0.20 ml. of 0.15 M sodium azide, 0.050 ml. of 4.8×10^{-2} M GSH, an appropriate volume of the peroxidase solution, sufficient 0.15 M NaCl to make the final volume of each sample 3.00 ml., and 0.10 ml. of 2.4×10^{-2} M ascorbic acid. The samples were incubated at 37° in test tubes without shaking. After 90 minutes of incubation, the reaction was stopped by treating the sample with carbon monoxide, and choleglobin was determined as described previously (1, 2). For a single quantitative assay, samples containing four to six different concentrations of enzyme were prepared in addition to a control sample which contained no enzyme. A blank sample, containing only hemoglobin and saline, was analyzed without incubation. The choleglobin concentration in these samples was then plotted against the enzyme concentration. The amount of enzyme producing a 50 per cent inhibition in choleglobin formation under these conditions has arbitrarily been designated to represent 1 unit of enzyme activity. Assay procedure 1 may be used for the qualitative assay of fractions from a column by using the percentage of inhibition of choleglobin formation as an indication of the presence of GSH peroxidase in the sample. In assays of this type, it is convenient

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[1] The abbreviations used are: EDTA, ethylenediaminetetraacetate; DEAE-cellulose, diethylaminoethyl cellulose.

to combine the hemoglobin, azide, GSH, and NaCl, so that only the enzyme solution and the ascorbic acid are added individually to each tube.

Peroxidase Assay Procedure 2—In this procedure, the rate of oxidation of GSH by hydrogen peroxide is used as a measure of peroxidase activity. GSH remaining in solution at a given time is determined by the mercuribenzoate procedure of Boyer (11). EDTA was utilized in the incubation mixture to reduce the nonenzymatic reaction rate to a low level. GSH-EDTA-buffer solution, 0.5 ml., (4.0×10^{-3} M GSH and 4.0×10^{-4} M EDTA in a 0.40 M phosphate buffer, pH 7.0) was added to a test tube containing an appropriate volume of the peroxidase solution and sufficient water to bring the volume to 4.50 ml. After a preincubation period of 5 minutes at 37°, the reaction was started by adding to each sample 0.50 ml. of 2.0×10^{-3} M H_2O_2 (warmed to 37°). The samples were incubated at 37° without shaking and at three minute intervals, aliquots were removed and analyzed for GSH in the following manner. A 1.00-ml. aliquot of the sample was added to a test tube containing 2.00 ml. of 1 M acetate buffer (pH 4.6), 1.00 ml. of 0.010 M $MgCl_2$, and 1.00 ml. of 6.0×10^{-4} M *p*-mercuribenzoate. The mercuribenzoate and acetate buffer stop the reaction of hydrogen peroxide and GSH and the latter is converted to the glutathione-mercuribenzoate complex. The presence of hydrogen peroxide in the sample does not affect the stability of the complex and optical density readings on the sample may be taken at any time between 15 minutes and 2 hours. The amount of GSH may be readily calculated from the optical density reading of the sample at 255 mμ. The presence of $MgCl_2$ in the sample prevents EDTA from interfering in the GSH determination. The "zero" time GSH concentration is determined in the same manner using an aliquot from a sample treated similarly but containing water in place of hydrogen peroxide. A blank sample containing all of the above reagents but no peroxidase is always run simultaneously in order to determine the rate of the nonenzymatic reaction of GSH and hydrogen peroxide. The GSH concentration in the incubated sample after 3, 6, and 9 minutes is plotted as shown in Fig. 1, and a straight line is drawn through the points. The enzyme activity is then determined using the equation of Tsuboi, *et al.* (12): activity $= 10 (\log C_0 - \log C_6)$. C_0 is the initial GSH concentration and C_6 is the GSH concentration after 6 minutes as determined from the graph. The activity of the blank sample is subtracted in order to determine the enzyme-catalyzed activity. This type of assay may be utilized to determine enzyme activity in the later stages of a purification procedure.[2] With purified enzyme preparations the absorption at 255 mμ due to protein is negligible. With impure preparations, the presence of inactive proteins or hemoglobin would necessitate a preliminary deproteinization in order to determine GSH and thus make this procedure much too difficult for routine use. Therefore, with impure enzyme preparations, assay procedure 1 is much easier to use.

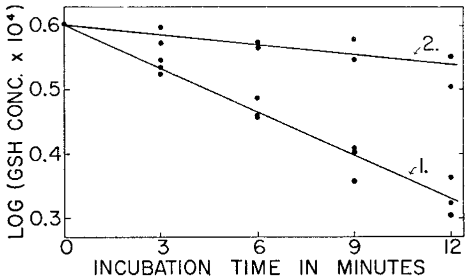

FIG. 1. The reaction of H_2O_2 and GSH in the presence of GSH peroxidase. The experiment was carried out as described in the text (assay procedure 2). *Curve 1*, with 0.10 ml. of the F_2 peroxidase preparation per sample; *Curve 2*, with no peroxidase.

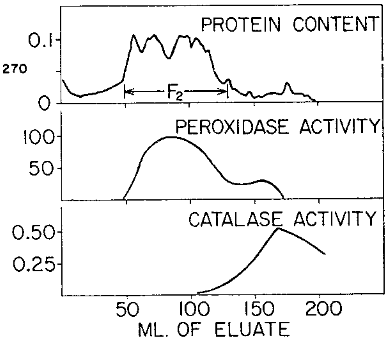

FIG. 2. Rechromatography of F_1 enzyme fraction on a DEAE-cellulose column. The experimental conditions are described in the text. Assay procedure 1 was used as a qualitative measure of peroxidase activity which is expressed in terms of percentage of inhibition of choleglobin formation. A 1.10 ml. aliquot of a 1:20 dilution of the eluent fractions was used for each assay. The optical density readings at 270 mμ were also made on 1:20 dilutions of the eluent fractions. Catalase activity is expressed in terms of m.eq. of perborate destroyed per ml. of a 1:20 dilution of the eluent fractions. The eluent fractions within the markers were pooled and quantitatively assayed for peroxidase activity (F_2 enzyme fraction).

RESULTS AND DISCUSSION

Purification of GSH Peroxidase on DEAE-cellulose Column—An erythrocyte enzyme preparation as used previously (1) was prepared from 1800 ml. of bovine blood by the acetone precipitation procedure[3] of Adams and Smith (13). The acetone precipitate was recovered by centrifugation, dissolved in water, frozen and stored in the Deep Freeze until used.

The frozen enzyme preparation was thawed and denatured proteins were removed by centrifugation. Phosphate buffer, pH 7.0, was added to the supernatant fluid to give a phosphate concentration of 0.005 M and the solution was run onto a DEAE-cellulose column (12.7 cm. $\times$ 7.1 cm.^{2}) prepared as described by Sober, *et al.* (14). Gradient elution was started using a 0.250 M phosphate buffer, pH 7.0, in the upper reservoir and 500 ml. of 0.005 M phosphate buffer, pH 7.0, in the mixing reservoir. The elution was carried out at room temperature and an automatic fraction collector was used to collect fractions. The column flow rate was approximtely 1.5 ml. per minute and fractions containing approximately 14 ml. each were collected. The fractions were assayed qualitatively for GSH peroxidase and quantitatively for protein content and catalase activity. The peroxidase activity appeared in the eluent fractions (110 to 290 ml.) slightly ahead of the catalase activity (210 to 450 ml.). The fractions containing peroxidase activity were combined (Fraction F_1) and quantitatively assayed for peroxidase activity. They were dialyzed against a 0.005 M phosphate buffer, pH

[2] Assays of the Fraction F_2 enzyme preparation indicate that one unit of activity by assay procedure 1 is equivalent to approximately 1.5 units by assay procedure 2.

[3] The procedure of Adams and Smith was slightly modified by omitting the pH adjustment step before the addition of acetone.

TABLE I
Summary of GSH peroxidase purification procedure

Fraction	Total peroxidase activity*	Protein	Specific activity	
			peroxidase	catalase†
	units	*mg.*	*units/mg.*	*m.eq./mg.*
Hemoglobin-free, protein precipitate	56,000	1300	43	
F_1 eluate from column	76,000	205	370	4.2
F_2 eluate from column	34,400	126	274	0.09

* As determined by assay procedure 1.
† Expressed as m.eq. of perborate destroyed per mg. of protein.

TABLE II
Effects of various sulfhydryl compounds on choleglobin formation in presence or absence of peroxidase

The experiment was carried out essentially as described for assay procedure 1 with azide omitted and the designated sulfhydryl compound replacing the GSH. The volume of each reagent was increased proportionately in order to provide a sufficient volume (10 ml.) in each sample for analyses at different time intervals. Final concentrations of the major components in the incubated samples were as follows: hemoglobin (crystalline), 0.10 gm. per 100 ml.; sulfhydryl compound, 4.0×10^{-4} M; and ascorbic acid, 8.0×10^{-4} M. F_2 GSH peroxidase preparation (0.80 ml.) was added to the samples marked with a plus sign.

Sulfhydryl compound	GSH peroxidase	Choleglobin formed in	
		30 min.	90 min.
		%	%
GSH	+	1.8	3.9
GSH	−	4.9	12.8
Cysteine	+	10.3	16.4
Cysteine	−	10.6	21.9
Ergothioneine	+	3.6	9.3
Ergothioneine	−	4.0	10.2
Cysteinylglycine	+	9.9	15.3
Cysteinylglycine	−	10.7	18.1
Thiomalic acid	+	8.0	17.6
Thiomalic acid	−	9.6	23.9
None	+	4.1	9.4
None	−	4.2	10.4

7.0, and were subsequently run onto a second DEAE-cellulose column (23 cm. × 0.64 cm.²). The gradient elution was again carried out at pH 7.0 with 500 ml. of 0.005 M buffer in the mixing reservoir and 0.250 M buffer in the upper reservoir, with a flow rate of approximately 8 ml. per hour. Fig. 2 correlates the protein content of the various eluent fractions (approximately 3 ml. per fraction) with the peroxidase and catalase activities.

The purification procedures and the degree of purification achieved in each step are shown in Table I. Although the specific activity of the enzyme preparation eluted from the second column (Fraction F_2) was less than that of the enzyme preparation which was put on the column, all of the catalase activity was removed in this purification step.

Further purification procedures were attempted using carboxymethylcellulose columns (8, 14), but were unsuccessful since the protein was not retained on the column at either pH 6.5 or 5.5 using a 0.005 M phosphate buffer.

At the present state of purification the absorption spectra of individual eluent fractions with the highest peroxidase specific activity show no indication of a Soret band at 400 to 420 mμ. The Soret band of a heme protein eluted just before the peroxidase and the Soret band of catalase in the fractions eluted after the peroxidase are discernible. Consequently, GSH peroxidase is probably not a heme enzyme. This is in accord with the results of the previous work (1, 2) where it was shown that the enzyme was not inhibited by cyanide or azide, both of which inhibit heme enzymes.

Because of the instability of purified GSH peroxidase,[4] no further purification procedures were attempted. Fraction F_2 was divided into 20-ml. portions and stored in the Deep Freeze. These portions, when thawed, had an initial activity of approximately 40 units per mg. of protein (assay procedure 2). Due to loss of enzymatic activity during storage in the refrigerator, the enzyme preparation as used in the various experiments had an activity ranging from 7 to 20 units per mg.

Optimal pH of GSH Peroxidase—Measurement of the decrease in GSH concentration in the presence of hydrogen peroxide (assay procedure 2) was utilized in the determination of the optimal pH for GSH peroxidase activity. In these experiments the rate of oxidation of GSH by hydrogen peroxide was determined in phosphate buffers in the presence (+) or absence (−) of the peroxidase preparation. The rates, expressed as μmoles of GSH oxidized per ml. per minute, were as follows: pH 6.0, (+) 0.003, (−) 0.002; pH 6.5, (+) 0.008, (−) 0.002; pH 7.0, (+) 0.017, (−) 0.006; pH 7.5, (+) 0.030, (−) 0.015; pH 8.0, (+) 0.084, (−) 0.040; pH 8.8, (+) 0.120, (−) 0.067. The rapid increase in the rate of the nonenzymatic reaction above pH 7.5 makes it difficult to ascertain accurately the effect of the enzyme above pH 8.0, but the results indicate that the rate of the enzyme-catalyzed reaction is highest at a pH of 8 or above. It is significant that GSH peroxidase shows no activity at a pH of 6 or below in contrast to various plant peroxidases which show optimal activity at pH 4.5 to 5.5 (15). A pH of 7.0 has been utilized in the experiments described in the following sections since the rate of the nonenzymatic reaction of GSH and hydrogen peroxide is low at pH 7.0.

Effect of Various Sulfhydryl Compounds in Protecting Hemoglobin from Oxidative Breakdown—Since one of the primary functions of GSH peroxidase in erythrocytes appears to be the protection of hemoglobin from oxidative breakdown, it is of interest to know whether other sulfhydryl compounds might serve with peroxidase in this protective function. In Table II are shown the results of adding GSH, cysteine, ergothioneine, cysteinylglycine, and thiomalic acid, respectively, to the ascorbic acid-hemoglobin system in the presence or absence of GSH peroxidase. The results of the experiment are complicated by the fact that some of these compounds will serve as hydrogen donors for the coupled oxidation of hemoglobin thus tending to increase the choleglobin formation beyond that produced by ascorbic acid alone. The results with GSH are in accord with those described previously (1). In the absence of peroxidase, GSH causes a slight increase in choleglobin formation while in

[4] Attempts thus far to protect the enzyme from this loss of activity have proven unsuccessful. The presence of EDTA, GSH, or crystalline bovine serum albumin in separate samples of the F_2 peroxidase preparation did not significantly change the rate at which the enzyme activity was lost when stored in the refrigerator.

the presence of peroxidase the amount of choleglobin is markedly reduced. Cysteine, cysteinylglycine, and thiomalate undergo a coupled oxidation with oxyhemoglobin since the choleglobin formation in samples containing these sulfhydryl compounds is approximately twice that of the control sample. There is some indication that the presence of peroxidase in these samples causes a reduction in the amount of choleglobin produced, but these differences are of questionable significance. When compared with the control sample, the over-all effect of cysteine, cysteinylglycine, and thiomalate in the presence of GSH peroxidase is to increase the choleglobin formation. Thus, these compounds do not protect hemoglobin from oxidative breakdown. The results with ergothioneine give no indication that it might serve either as a hydrogen donor in the coupled oxidation of hemoglobin or as a hydrogen donor in the peroxidatic destruction of hydrogen peroxide. Thus, ergothioneine, a normal constituent of erythrocytes, does not bring about oxidative destruction of hemoglobin nor does it protect hemoglobin from oxidative breakdown.

Effect of Various Hydrogen Donors in Peroxidatic Breakdown of Hydrogen Peroxide—Assay procedure 2 was used to study the specificity of GSH peroxidase in regard to sulfhydryl hydrogen donors. In these experiments, the rate of oxidation of the sulfhydryl compound by hydrogen peroxide at pH 7.0 was determined in the presence (+) or absence (−) of the peroxidase preparation. The rates, expressed as μmoles of —SH oxidized per ml. per minute, were as follows: GSH, (+) 0.019, (−) 0.005; cysteine, (+) 0.065, (−) 0.044; and cysteinylglycine, (+) 0.027, (−) 0.023. With the exception of GSH, only cysteine showed a significant increase in the rate of oxidation of sulfhydryl groups in the presence of peroxidase. The nonenzymatic rate of oxidation of sulfhydryl groups was high in the case of both cysteine and cysteinylglycine. The presence of peroxidase in samples containing cysteinylglycine brought about a slight increase in the rate of oxidation of sulfhydryl groups, but the increase is of doubtful significance. Thiomalate was not oxidized in significant amounts either with or without peroxidase under the conditions of this experiment.

The results of this experiment are thus essentially in accord with the results of the previous experiment in which the effect of these sulfhydryl compounds in the protection of hemoglobin from oxidation was studied. Although the rate of oxidation of cysteine is increased when the enzyme is present, the high rate of the nonenzymatic reaction would indicate that the enzyme is not necessary for the reaction to proceed. GSH has a low nonenzymatic rate of oxidation by hydrogen peroxide, and in the presence of peroxidase the rate of oxidation is markedly increased. Consequently, these results are in accord with the view that GSH is the normal hydrogen donor for this reaction inside erythrocytes.

In additional experiments, GSH peroxidase was tested qualitatively for activity in catalyzing the breakdown of hydrogen peroxide with either *o*-tolidine, pyrogallol, or guaiacol serving as a hydrogen donor (5). At a pH of 7 under conditions where 6 μg. of horse-radish peroxidase produced a marked color change within 30 seconds, 1 ml. of Fraction F_2 GSH peroxidase preparation produced no change in the color of the sample in 30 minutes. When tested for activity with GSH as a hydrogen donor (assay procedure 2), 30 μg. of horse-radish peroxidase showed no activity while the Fraction F_2 peroxidase preparation had an activity of approximately 20 units per ml. Thus, GSH peroxidase does not catalyze the breakdown of hydrogen peroxide in the presence of *o*-tolidine, guaiacol, or pyrogallol, even though these compounds are usually used as hydrogen donors in tests for peroxidase activity.

Utilization of Peroxidase-GSH System in Protection of Hemoglobin from Oxidative Breakdown—The effectiveness of the peroxidase-GSH system in protecting hemoglobin from oxidative breakdown produced by either ascorbic acid or hydrogen peroxide has been shown previously (1, 2). A number of additional compounds, such as epinephrine (16), phenylhydrazine (17, 18), and acetylphenylhydrazine (19) also react with oxyhemoglobin and lead to a partial destruction of hemoglobin. Certain drugs, such as primaquine, acetanilide, and sulfanilamide, which have hemolytic effects *in vivo* in drug-sensitive individuals, appear to react in a manner similar to that of phenylhydrazine and acetylphenylhydrazine (20). Consequently, it is of importance to know the means by which hemoglobin of erythrocytes is protected from the deleterious effects of these drugs or their metabolites.

In the present studies, epinephrine, acetylphenylhydrazine, and cysteine have been utilized to bring about the oxidative breakdown of hemoglobin. The products of hemoglobin breakdown under these conditions have not been fully characterized. In the reaction of epinephrine and oxyhemoglobin, the product which is formed appears to be identical with choleglobin, since its reduced carbon monoxide derivative shows a sharp absorption band at 627 to 630 mμ. The reaction of cysteine with oxyhemoglobin yields a product which appears to be predominantly choleglobin, but small amounts of other products may also be present. Acetylphenylhydrazine undoubtedly leads to a mixture of oxidative breakdown products of hemoglobin since these samples, after treatment with carbon monoxide and dithionite, show a general increase in absorption from 600 to 635 mμ, but do not show the distinctive band of carboxycholeglobin at 627

TABLE III

Protection of hemoglobin from oxidative breakdown by peroxidase-GSH system

The experiment was carried out essentially as described for assay procedure 1 with azide omitted and the designated compound replacing ascorbic acid. The volume of each reagent was increased proportionately in order to provide a sufficient volume (10 ml.) in each sample for analyses at different time intervals. Each incubated sample contained hemoglobin (crystalline) 0.10 gm. per 100 ml., and GSH, 4.0×10^{-4} M. The other compounds were added as designated to give the following concentrations in the incubated samples: cysteine, 1.6×10^{-3} M; epinephrine, 5.5×10^{-4} M; and acetylphenylhydrazine, 6.7×10^{-4} M. F_2 GSH peroxidase preparation (0.50 ml.) was added to the samples marked with a plus sign. The optical density at 627 mμ at zero time in these samples was 0.019.

Sample	GSH peroxidase	Optical density of sample at 627 mμ	
		30 min.	90 min.
Cysteine	+	0.075	0.101
Cysteine	−	0.106	0.134
Epinephrine	+	0.045	0.056
Epinephrine	−	0.063	0.091
Acetylphenylhydrazine	+	0.029	0.043
Acetylphenylhydrazine	−	0.068	0.103

to 630 mμ. In order to determine the effect of the peroxidase-GSH system, it was necessary to establish conditions whereby oxidative breakdown of hemoglobin was submaximal. Under these conditions any increase or decrease in the rate of breakdown of hemoglobin would be readily apparent. The conditions utilized in the experiments described in Table III were found to bring about a submaximal rate of hemoglobin breakdown.

In Table III are shown the results of these studies on the oxidative breakdown of hemoglobin in the presence or absence of GSH peroxidase and GSH. In this experiment, the increase in optical density at 627 mμ after treatment of the sample with carbon monoxide and dithionite is used as a measure of hemoglobin breakdown. It will be seen that the peroxidase-GSH system is effective in protecting hemoglobin from oxidative breakdown which is produced by either cysteine, epinephrine or acetylphenylhydrazine. It may be inferred that the peroxidase-GSH system is also of importance *in vivo* in protecting hemoglobin from the deleterious effects of certain drugs and their metabolites.

SUMMARY

Column chromatography with a modified cellulose anion exchange column (diethylaminoethyl cellulose) has been utilized for purification of the glutathione peroxidase of bovine erythrocytes. By using this purification procedure, the peroxidase activity of erythrocytes has been completely separated from catalase activity. Two distinct assay procedures for determination of glutathione peroxidase activity are also described.

The hydrogen donor specificity of glutathione peroxidase of erythrocytes has been investigated. Reduced glutathione serves as the most effective hydrogen donor for the peroxidatic destruction of hydrogen peroxide as catalyzed by glutathione peroxidase. Pyrogallol, guaiacol, and *o*-tolidine will not serve as hydrogen donors in the peroxidatic breakdown of hydrogen peroxide as catalyzed by glutathione peroxidase.

Glutathione peroxidase has optimal activity at a pH of 8 or above and shows no significant enzymatic activity at a pH of 6 or below.

The effectiveness of the peroxidase-glutathione system in protecting hemoglobin from oxidative breakdown has also been studied. This system will protect hemoglobin from oxidative breakdown produced by the coupled oxidation of hemoglobin with either cysteine, epinephrine, or acetylphenylhydrazine.

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